Nau mai, welcome to

Applied Engineering Optimisation

ENEL445

Weeks 4+: Constrained optimisation

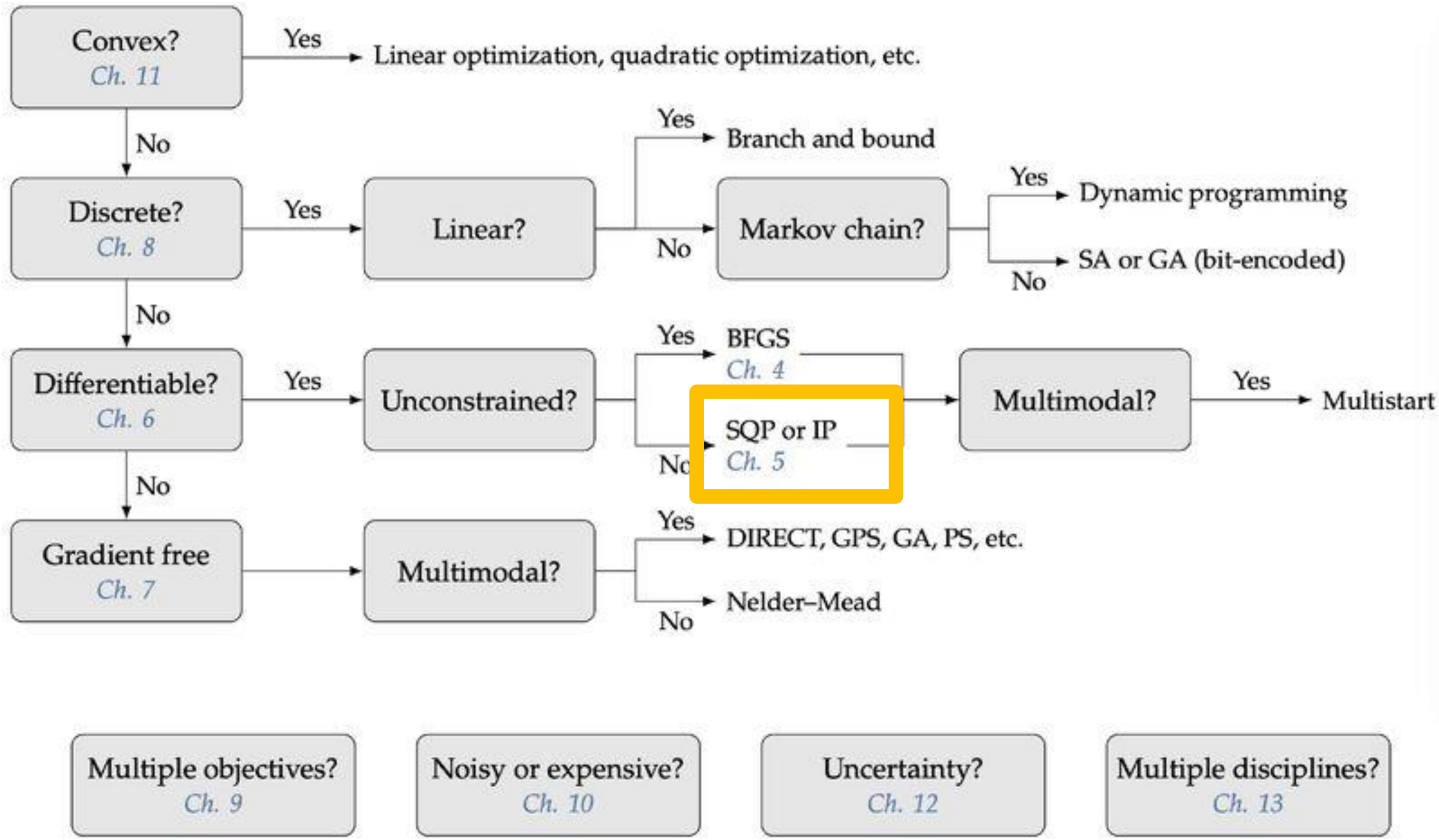
- This is a *difficult* topic.
- The course reader may be helpful, but please do **attend the lectures** and **stop me and ask questions** if you are not following the lecture material.

Topics

We will cover:

- Optimality conditions
  - Equality constraints
  - Inequality constraints
  - KKT conditions
- Algorithms for solving constrained problems
  - Penalty methods
  - Sequential quadratic programming (SQP)
  - Interior point
  - Duality

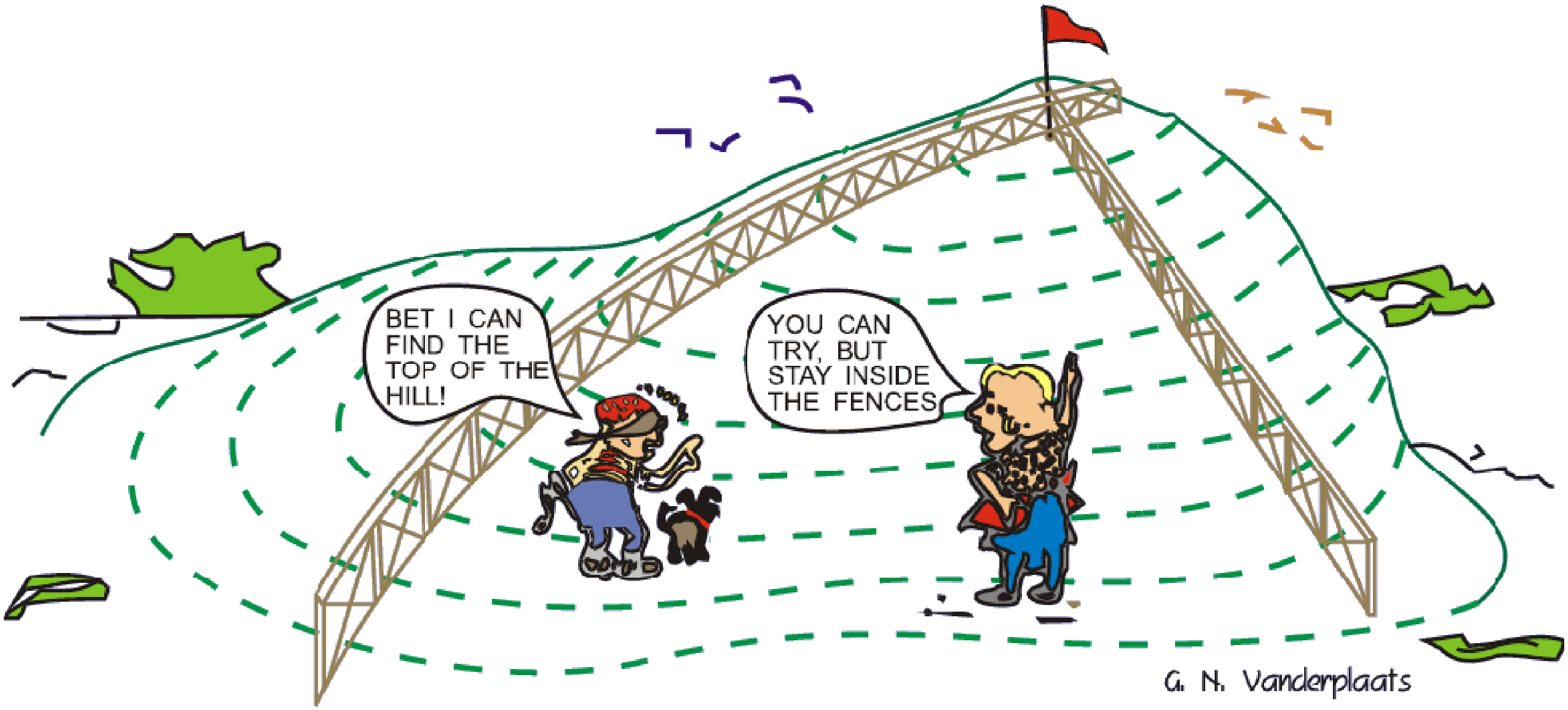
Which optimiser should I use?



Constrained optimisation

- Most optimisation problems are constrained
- Most variables (and basically all physical variables) have limits, e.g.
  - if you are optimising the power system (something that happens every half hour in NZ), each generator and each line will have limits.
  - if optimising an electrical appliance, there will be budget, dimension, power usage constraints, etc.

Constrained optimisation



Definition

$$
\begin{aligned}
\min\ & f(\mathbf{x}) \\
\text{subject to}\ & \underline{x}_i \le x_i \le \bar{x}_i, i = 1, 2, ..., n \\
& g_j(\mathbf{x}) \le 0,\ \ j = 1, 2, ..., n_g \\
& h_l(\mathbf{x}) = 0, l = 1, 2, ..., n_h.
\end{aligned}
$$

Equality constraints

Let's start by considering the case where we only have equality constraints, i.e. no inequality constraints of any sort.

Recap from week 3. A necessary condition for optimality was: $\nabla f(\mathbf{x}^*)^T \mathbf{p} \geq 0$

(for all ***p***) i.e. in all directions the objective function does not improve.

Equality constraints

What if we have equality constraints? It is now okay for the objective function to improve in some direction ***p*** so long as that direction ***p*** is not feasible, i.e. we cannot go there.

So we need $\nabla f(\mathbf{x}^*)^T \mathbf{p} \geq 0$ but only for feasible directions ***p***, rather than for all ***p***.

How to find feasible directions

- Taylor-series:

$$h_j(\mathbf{x}+\mathbf{p}) \approx h_j(\mathbf{x}) + \nabla h_j(\mathbf{x})^T \mathbf{p}, l = 1, 2, ..., n_h.$$

- Assuming *x* is feasible, then $h_j(\mathbf{x}) = 0$

$$h_j(\mathbf{x}+\mathbf{p}) \approx \nabla h_j(\mathbf{x})^T \mathbf{p}, l = 1, 2, ..., n_h.$$

- To remain feasible,

$$h_j(\mathbf{x}+\mathbf{p}) = 0 \text{ for all } j$$

- We can write this in matrix form using the Jacobian matrix

$$J_h(\mathbf{x})\mathbf{p} = 0$$

Optimality conditions

- For x^* to be a constrained optimum

$$\nabla f(\mathbf{x}^*)^T \mathbf{p} \geq 0 \text{ for all } \mathbf{p} \text{ such that } J_h(\mathbf{x}^*)\mathbf{p} = 0.$$

- Now, $h_j(x) = 0$ is a hyper-plane in *N* space.
- So, if ***p*** is a feasible direction, so is –***p***. Thus

$$\nabla f(\mathbf{x}^*)^T \mathbf{p} = 0 \text{ for all } \mathbf{p} \text{ such that } J_h(\mathbf{x}^*)\mathbf{p} = 0.$$

($\nabla f(\mathbf{x}^*)^T \mathbf{p}$ greater than zero implies negativity when ***p*** is replaced by –***p***)

Lagrange multipliers

- We now have two equations:

$$\nabla f(\mathbf{x}^*)^T p = 0$$
$$J_h(\mathbf{x}^*) p = 0$$

- The rows of J_h are gradients of the constraints, so the objective function gradient must be a linear combination of the gradients of the constraints.

$$\nabla f(\mathbf{x}^*) = -\sum_{j=1}^{n_h} \lambda_j \nabla h_j(\mathbf{x}^*)$$

Lagrangian function

- In matrix form:

$$\nabla f(\mathbf{x}) = -J_h(\mathbf{x})\lambda$$
$$h(\mathbf{x}) = 0$$

- It is sometimes convenient to define the Lagrangian function

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + h(\mathbf{x})^T \lambda.$$

- Taking the gradient of this w.r.t. **x** and λ yields the matrix conditions above, so we could try to find the critical point(s) of the Lagrangian via unconstrained optimisation – but not usually the best way.

Second-order conditions

- The conditions described are first-order (i.e. no second derivatives) and are necessary but not sufficient.
- To make sure the point is a constrained minimum, we need to check the Hessian function (similar to unconstrained optimisation).
- Recall that for unconstrained optimisation the Hessian had to be positive definite, i.e.

$$\mathbf{p}^T \mathbf{H}_f(\mathbf{x}^*)\mathbf{p} > 0.$$

Second-order conditions

- As before, we only need to check directions p which are feasible.

Example 5.2

Example 5.3

Inequality constraints

$$\begin{aligned} \min\ & f(\mathbf{x}) \\ \text{subject to}\ & \underline{x}_i \leq x_i \leq \bar{x}_i, i = 1, 2, ..., n \\ & g_j(\mathbf{x}) \leq 0,\ \ j = 1, 2, ..., n_g \\ & h_l(\mathbf{x}) = 0, l = 1, 2, ..., n_h. \end{aligned}$$

Important terminology

- A constraint is *active* if $g_j(\mathbf{x}^*) = 0$.
- A constraint is *inactive* if $g_j(\mathbf{x}^*) < 0$.
- A constraint is *feasible* when $g_j(\mathbf{x}^*) \leq 0$.

Optimality conditions

- As before, if x^* is an optimum, any small enough feasible step **p** from the optimum must result in a function increase. Based on Taylor series expansion, we had the condition:

$$\nabla f(\mathbf{x}^*)^T \mathbf{p} \geq 0$$

- To consider inequality constraints, we use linearization as before:

$$g_j(\mathbf{x} + \mathbf{p}) \approx g_j(\mathbf{x}) + \nabla g_j(\mathbf{x})^T \mathbf{p} \leq 0, j = 1, 2, ..., n_g.$$

Optimality conditions

- There are two possibilities for each constraint: whether the constraint is inactive $g_j(\mathbf{x}^*) < 0$ or active $g_j(\mathbf{x}^*) = 0$.
- If a given constraint is inactive, we do not need to add any condition for it because we can take a step ***p*** in any direction and remain feasible as long as the step is small enough.
- Thus, we only need to consider the active constraints for the optimality conditions.

Active inequality constraints

- If constraint *j* is active $(g_j(\mathbf{x}) = 0)$ then the nearby point $g_j(\mathbf{x} + \mathbf{p})$ is only feasible if $\nabla g_j(\mathbf{x})^T \mathbf{p} \leq 0$ for all constraints *j* that are active.

- In matrix form: $J_g(\mathbf{x})\mathbf{p} \leq 0$

where the Jacobian matrix includes only the gradients of the active constraints.

- We thus have two conditions

$$\nabla f(\mathbf{x}^*)^T \mathbf{p} \geq 0$$

$$J_g(\mathbf{x})\mathbf{p} \leq 0$$

Farkas' Lemma

- Farkas' Lemma states that given this, one of two possibilities occur:

1. There exists an $n \times 1$ vector $\mathbf{p}$ such that $J_g^T \mathbf{p} \leq 0$ and $\nabla f^T \mathbf{p} < 0$.

2. There exists an $n \times 1$ vector σ such that $J_g \sigma = \nabla f$ and $\sigma \geq 0$.

- In case 1, there is a feasible direction $J_g(\mathbf{x}^*)^T \mathbf{p} \leq 0$ in which we can descend $\nabla f(\mathbf{x}^*)^T \mathbf{p} < 0$.
- So, case 2 is our optimality condition.

$$\nabla f(\mathbf{x}^*) + J_g(\mathbf{x}^*)^T \sigma = 0, \text{with } \sigma \geq 0.$$

- See course reader or textbook for proof.

Lagrange multipliers

- Similar to the condition for equality constraints,

$$\nabla f(\mathbf{x}) = -J_h(\mathbf{x})\lambda$$
$$h(\mathbf{x}) = 0$$

σ corresponds to the Langrange multipliers for the inequality constraints and carries the additional restriction that they are non-negative.

$$\nabla f(\mathbf{x}^*) + J_g(\mathbf{x}^*)^T\sigma = 0, \text{with } \sigma \geq 0.$$

Slackness

- As in the equality constrained case, we can construct a Lagrangian function whose stationary points are candidates for optimal points.
- We need to include all inequality constraints in the optimality conditions because we do not know in advance which constraints are active.
- To represent inequality constraints in the Lagrangian, we replace them with the equality constraints defined by:

$$g_j + s_j^2 = 0, j = 1, ..., n_g.$$

- where s_j is called a *slack variable*.

Lagrangian

- The Lagrangian including both equality and inequality constraints is then

$$\mathcal{L}(\mathbf{x}, \lambda, \sigma, \mathbf{s}) = f(\mathbf{x}) + \lambda^T h(\mathbf{x}) + \sigma^T (g(\mathbf{x}) + \mathbf{s} \odot \mathbf{s})$$

where σ represents the Lagrange multipliers associated with the inequality constraints. We use $\odot$ to represent the element-wise multiplication of s.

KKT conditions

- Similar to the equality constrained case, we seek a stationary point for the Lagrangian, but now we have additional unknowns: the inequality Lagrange multipliers and the slack variables.
- Taking partial derivatives of the Lagrangian with respect to each set of unknowns and setting those derivatives to zero yields the first-order optimality conditions.

KKT conditions

KKT conditions

- In addition to the conditions for a stationary point of the Lagrangian, recall that we require the Lagrange multipliers for the active constraints to be nonnegative. Putting all these conditions together in matrix form, the first-order constrained optimality conditions are as follows:

$$\begin{aligned} \nabla_f + J_h^T \lambda + J_g^T \sigma &= 0 \\ h &= 0 \\ g + s \odot s &= 0 \\ \sigma \odot s &= 0 \\ \sigma &\geq 0 \end{aligned}$$

Second-order conditions

- As in the equality constrained case, these first-order conditions are necessary but not sufficient. The second-order sufficient conditions require that the Hessian of the Lagrangian must be positive definite in all feasible directions. That is,

$$\mathbf{p}^T H_\lambda \mathbf{p} > 0$$

$$J_h \mathbf{p} = 0$$

$$J_g \mathbf{p} \leq 0$$

Example 5.4

Example 5.5

Practical considerations

- Although these examples can be solved analytically, this is often difficult or impractical.
- The KKT conditions quickly become challenging to solve analytically and as the number of constraints increases, trying all combinations of active and inactive constraints becomes intractable.

Meaning of Lagrange multipliers

- The Lagrange multipliers quantify how much the corresponding constraints drive the design.
- More specifically, a Lagrange multiplier quantifies the sensitivity of the optimal objective function value $f(\mathbf{x}^*)$ to a variation in the value of the corresponding constraint.
- They quantify the relative importance of the constraints, e.g. when a constraint is inactive, the corresponding Lagrange multiplier is zero.

Algorithms for solving constrained optimization problems

- In general, if we can solve the KKT conditions, we have solved the optimization problem.
- However, as mentioned previous, there is a combinatorial issue with directly solving the KKT conditions for many problems, so we need an algorithm.
- Most important algorithms are *interior-point* and *SQP*.

Penalty methods

- The concept behind penalty methods is intuitive: to transform a constrained problem into an unconstrained one by adding a penalty to the objective function when constraints are violated or close to being violated.

$$\hat{f}(\mathbf{x}) = f(\mathbf{x}) + \mu\pi(\mathbf{x})$$

- Exterior vs interior penalty methods

Penalty methods

- We can use unconstrained optimization techniques to minimize $\hat{f}(\mathbf{x})$
- However, instead of just solving a single optimization problem, penalty methods usually solve a sequence of problems with different values of μ to get closer to the actual constrained minimum.
- As μ tends to infinity, we get the exact solution. However, we cannot start with a large value for μ because this results in the Hessian being ill-conditioned.
- Instead, we start with a small μ and gradually increase it, using the previous solution as the starting point.

Sequential Quadratic Programming

- Main idea:
  - solve the KKT conditions rather than the original problem
  - this gives a system of equations r(x,λ) = 0, which, using Newton's method, can be solved by via solving a sequence of linear systems.

$$\begin{bmatrix} H_{\mathcal{L}} & J_h^\top \\ J_h & 0 \end{bmatrix} \begin{bmatrix} p_x \\ p_\lambda \end{bmatrix} = \begin{bmatrix} -\nabla_x \mathcal{L} \\ -h \end{bmatrix}$$

SQP

- In turns out this is equivalent to solving a local quadratic problem (QP) approximation

$$\begin{aligned} \underset{p}{\text{minimize}} \quad & \frac{1}{2} p^\intercal H_{\mathcal{L}} p + \nabla f^\intercal p \\ \text{subject to} \quad & J_h p + h = 0 \,. \end{aligned}$$

- In practice, SQP requires the Hessian -> quasi-Newton approximation used e.g. BFGS formula, with line search, similarly to last week*.

* However, things are slightly more complicated in line search as not only are we trying to reduce the objective function, we are also trying to “steer away” from infeasibility.

Interior point

- These methods form an objective similar to the interior penalty but with the key difference that instead of penalizing the constraints directly, they add slack variables to the set of optimization variables and penalize the slack variables.

$$\min_{x,s} \; f(\mathbf{x}) - \mu_b \sum_{j=1}^{n_g} \log s_j$$

$$\text{subject to} \quad g(\mathbf{x}) + s = 0$$

$$h(\mathbf{x}) = 0$$

Interior point

- Similar to SQP, Newton’s method is used to solve KKT conditions.
- However, instead of solving the KKT conditions of the original problem, we solve the KKT conditions of the interior-point formulation.

Interior point

- If $\mu_b = 0$, then the interior-point formulation is equivalent to the original constrained problem.
- Similarly to penalty methods, we have to move slowly to prevent issues with conditioning due to the gradients.

$$\min_{x,s}\ f(\mathbf{x}) - \mu_b \sum_{j=1}^{n_g} \log s_j$$

$$\begin{aligned} \text{subject to}\quad & g(\mathbf{x}) + s = 0 \\ & h(\mathbf{x}) = 0 \end{aligned}$$

Interior point

- Determining the correct step-size (line search again!) is a crucial part of any interior-point algorithm, and it should be noted that this is used to enforce $s > 0$ implicitly.
- Hence, at each iteration, the parameters μ_b, x, and s are updated until we converge.

$$\min_{x,s} \; f(\mathbf{x}) - \mu_b \sum_{j=1}^{n_g} \log s_j$$

$$\text{subject to} \quad g(\mathbf{x}) + s = 0$$

$$h(\mathbf{x}) = 0$$

Duality

- We have shown a *barrier function* interior point algorithm.
- In practice, the most successful algorithms are primal-dual ones, which are based on the concept of the *Lagrangian dual function*.
- Why?
  - The dual problem is always a convex problem
  - The dual problem is unconstrained
  - If certain (fairly common) conditions are satisfied, the solution of the dual problem is equal to the original (“primal”) problem, i.e. zero duality gap.

Duality derivation

Duality example 1

$$\min_x (x-2)^2$$
$$\text{subject to } x \leq 1$$

Duality example 2

$$\min_x x^4 - 50x^2 + 100x$$
$$\text{subject to } x \geq -4.5$$

Duality example 2